

Coding Challenge: Cloud Deployment & CI/CD(120 mins)

Scenario:

You are working as a **DevOps Engineer** at **TechSphere Inc.**, a company building a React-based **Contact Management System (CMS)**. The backend is developed using **Node.js with Express and MongoDB**, exposing a **RESTful API**.

The company is transitioning to **Cloud Deployment** with **CI/CD pipelines**. Your task is to **deploy the API and React frontend on the cloud**, set up a **CI/CD pipeline using GitHub Actions**, and ensure **automated testing**.

Task Overview

1 Set Up a Cloud Deployment (40 mins)

- **Deploy the API on a Cloud Platform** (AWS/GCP/Azure).
- Use **Docker** to containerize the API.
- Use a **Managed Database Service** (MongoDB Atlas or AWS DynamoDB).
- Deploy the **React frontend** on a cloud service (Vercel/Netlify/S3+CloudFront).

✓ **Expected Outcome:** API and frontend are accessible via a cloud URL.

2 Configure CI/CD with GitHub Actions (30 mins)

- **Create a GitHub Actions Workflow** that:
 - Runs tests on **every push and PR**.
 - Deploys the API and frontend **on merge to main**.
- Use a **CI/CD pipeline** to automate deployment.

✓ **Expected Outcome:**

- On each commit, tests run automatically.
 - On merging to main, deployment is triggered.
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3 Implement Automated Testing (30 mins)

- **Write Unit Tests for API** using Jest.

- **Write Integration Tests** to validate API responses.
- Ensure tests run **before deployment** in CI/CD.

✓ **Expected Outcome:**

- Running `npm test` in API repo executes all test cases.
- CI/CD pipeline blocks deployment on failed tests.

4 Version Control & Collaboration (20 mins)

- Use **Git branching strategy** (Feature branching, PRs, Code reviews).
- Merge using **Git rebase** or **Git merge**.
- Use **GitHub Issues** to track progress.

✓ **Expected Outcome:**

- Proper Git branch structure.
- Successful feature merging without conflicts.



Submission Guidelines

📌 **Deliverables:**

- **Cloud URLs** for API and Frontend.
- GitHub repo with **CI/CD config** (`.github/workflows/ci-cd.yml`).
- Test results in **CI/CD logs**.
- Screenshots of **successful deployment and test cases**.



Self-Evaluation Criteria

Criteria	Below Expectation	Satisfactory	Excellent
Cloud Deployment	Not deployed or issues in accessibility	API and frontend deployed, but with minor issues	Fully deployed, properly accessible

CI/CD Pipeline	Not configured or failing	CI/CD set up, but missing test automation	CI/CD automates tests & deployment successfully
Automated Testing	No test coverage or failing tests	Basic unit tests added	Comprehensive unit & integration tests
Version Control	No branching strategy followed	Basic branching & merging used	Proper branching, rebasing, and PR reviews followed
